



HEALTHCARE DATA ANALYSIS PROJECT REPORT

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FOR: BLOOMY TECHNOLOGIES
DATA ANALYTICS COURSE

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Executive Summary

This report presents a comprehensive healthcare data analysis for Wellness Hospital based on 500 patient records across 4 regional branches and 5 medical departments. Across the analyzed period, Wellness Hospital generated \$141M in Total Revenue, maintaining an Average Treatment Cost of \$245K, an Average Length of Stay of 7.5 days, and an Average Patient Satisfaction score of 3.0 out of 5.0.

The primary objective of this project is to evaluate clinical operational performance, financial distribution, treatment efficacy, and physician metrics to provide data-driven recommendations to hospital executive management.

Data Preparation & Cleaning

In compliance with project specifications, the raw dataset underwent cleaning and validation prior to visualization in Power BI.

Missing Value & Duplicate Check: Records were checked for missing values, correctness and duplicate entries.

Text Harmonization: Categorical variables (Gender, State, Hospital Branch, Diagnosis, Treatment Type) were standardized to resolve spelling variations and inconsistent capitalization.

Data Type, Date & Metric Validation: Appropriate data types were used each column. Admission Date and Discharge Date formats were unified. The field Length of Stay was re-calculated to ensure accuracy.

Age Group Categorization: Patient ages were grouped into distinct demographics in two columns for Age Range and Age Group. e.g. Child: 0–17 years Young Adult: 18–35 years Adult: 36–60 years Senior: 61+ years

WELLNESS HOSPITAL HEALTHCARE PERFORMANCE

Total Patients

500

Total Revenue

\$141M

Avg Treatment Cost

\$245K

Avg Patient
Satisfaction

3.0

Avg Length of Stay

7.5

Executive Dashboard

Patient Analysis
Dashboard

Financial Dashboard

Treatment Dashboard

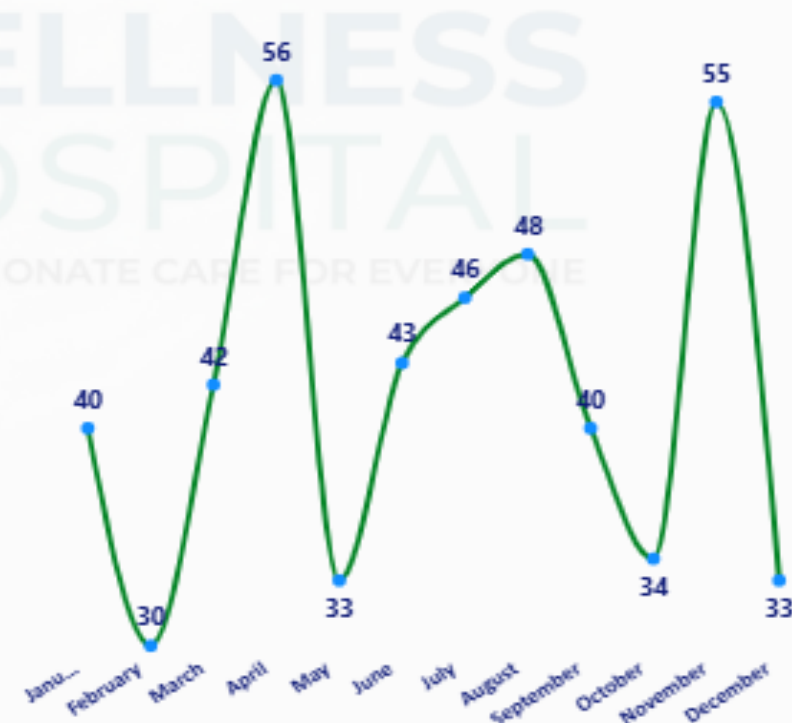
Doctor Performance
Dashboard

Analyzed by: Theresa Onyegiri
Date: September 2026

Monthly Revenue Trend



Monthly Admission Trend



Total Diagnoses

5

Total Doctors

5

Total Departments

5

Total Branches

4

Executive Dashboard Findings

- **Key High-Level Metrics:** Total Patients: 500, Total Revenue: \$141M, Avg Treatment Cost: \$245K , Avg Satisfaction: 3.0, Avg Length of Stay: 7.5 days.
- **Operational Scope:** 5 Departments, 5 Diagnoses, 5 Doctors, and 4 Hospital Branches.
- **Monthly Revenue & Admission Trends:** Revenue peaked in November at \$16.19M and March at \$14.91M, aligning with peak admission months, for November with 55 admissions and March with 56 admissions records. The lowest revenue occurred in February (\$8.13M) and October (\$9.49M).

WELLNESS HOSPITAL HEALTHCARE PERFORMANCE

Gender
☐ Female
☐ Male

Total Patients

500

Total Revenue

\$141M

Avg Treatment Cost

\$245K

Avg Patient
Satisfaction

3.0

Avg Length of Stay

7.5

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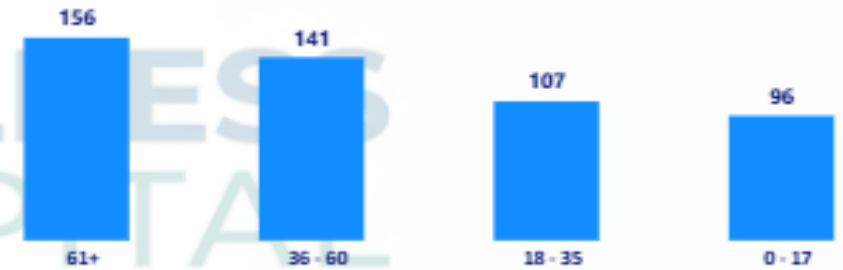
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Total Patients by Gender



Total Patients by Age Group



Total Patients by State



Total Patients by Branch



Patient Analysis Findings

- **Gender Breakdown:** Patients gender has a balanced distribution with 253 Males (50.6%) and 247 Females (49.4%).
- **Age Group Distribution:** Seniors (61+) represent the largest patient volume at 156 patients, followed by Adults (36–60) at 141, Young Adults (18–35) at 107, and Children (0–17) at 96.
- **Geographic & Branch Distribution:**
 - **Top Patient States:** FCT (106) and Lagos State (106), followed by Oyo (101), Rivers (98), and Kano (89).
 - **Top Branches by Volume:** Ibadan (131 patients) and Lagos (126 patients), followed by Port Harcourt (124) and Abuja (119).

WELLNESS HOSPITAL HEALTHCARE PERFORMANCE

Department

- ☐ Cardiology
- ☐ Neurology
- ☐ Oncology
- ☐ Orthopedics
- ☐ Pediatrics

Total Patients

500

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\$141M

Avg Treatment Cost

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Total Departments

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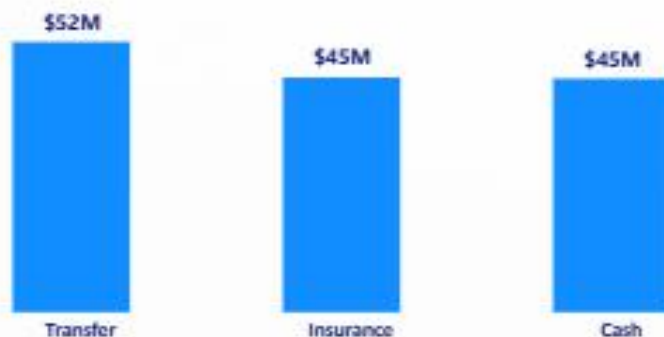
Total Revenue by Department



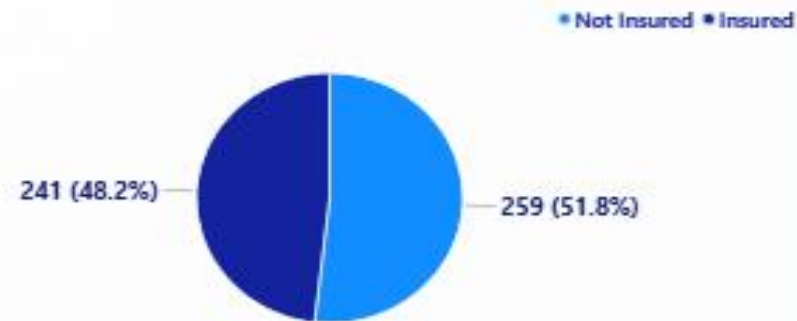
Total Revenue by Branch



Total Revenue by Payment Method



Total Patients by Insurance Status



Financial Analysis Findings

- **Departmental Revenue Leader:** Neurology generated the highest total revenue at \$33.3M, followed by Orthopedics (\$28.3M), Cardiology (\$28.2M), Oncology (\$28.0M), and Pediatrics (\$23.4M).
- **Branch Revenue Performance:** Ibadan Branch led overall revenue with \$40M, followed by Port Harcourt (\$35M), Lagos (\$33M), and Abuja (\$33M).
- **Payment Methods & Insurance:**
 - **Payment Channels:** Bank Transfer is the most prominent payment method generating \$52M, compared to Insurance (\$45M) and Cash (\$45M).
 - **Insurance Coverage:** 259 patients (51.8%) are insured, while 241 patients (48.2%) are uninsured.

WELLNESS HOSPITAL HEALTHCARE PERFORMANCE

Diagnosis

- Cancer
- Fracture
- Hypertension
- Malaria
- Migraine

Total Patients

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Total Revenue

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Avg Treatment Cost

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Total Departments

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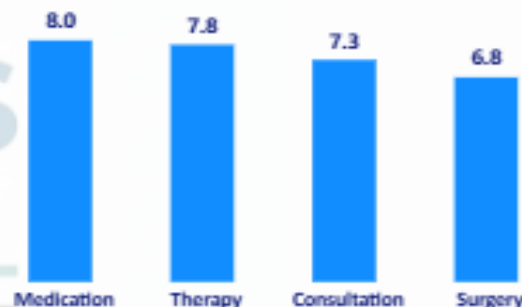
Total Patients by Diagnosis



Total Patients by Treatment Type



Avg Stay by Treatment Type



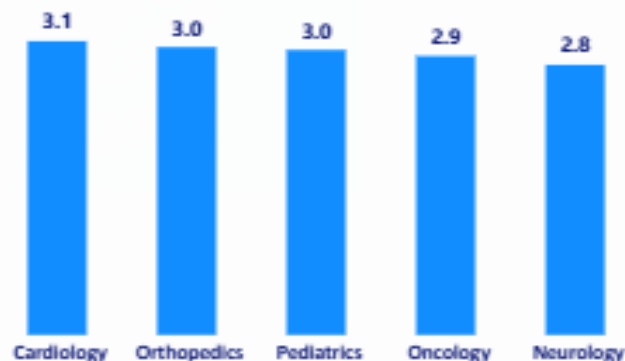
Readmission Rate by
Department



Readmission Rate by Diagnosis



Avg Patient Satisfaction by Department



Treatment Analysis Findings

➤ Diagnoses & Treatments:

- **Most Frequent Diagnosis:** Migraine leads with 111 patients, followed by Fracture (107), Cancer (103), Hypertension (103), and Malaria (76).
- **Most Frequent Treatment: Surgery** is the most common intervention (135 patients), followed by Medication (125), Consultation (121), and Therapy (119).

➤ Length of Stay by Treatment: Medication incurs the longest stay at 8.0 days, while Surgery has the shortest at 6.8 days.

➤ Readmission & Satisfaction Performance:

- **Departmental Readmissions: Cardiology** exhibits the highest readmission rate at 54.08%, followed by Orthopedics (50.00%).
- **Diagnosis Readmissions: Fracture** shows the highest rate at 51.40%, followed by Malaria (50.00%).
- **Departmental Satisfaction:** Cardiology achieved the highest satisfaction score (3.1/5.0), whereas Neurology scored lowest (2.8/5.0).

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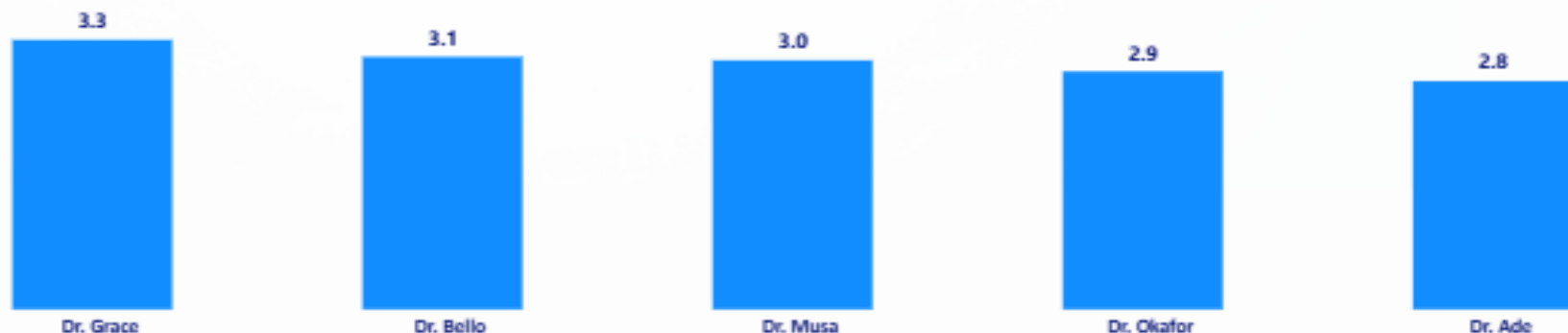
Total Patients by Doctor



Total Revenue by Doctor



Avg Patient Satisfaction by Doctor



Doctor Performance Analysis

Doctor Name	Patients Treated	Revenue Generated	Avg Patient Satisfaction
Dr. Musa	113	\$32.1M	3.0
Dr. Ade	111	\$32.7M	2.8
Dr. Bello	98	\$27.4M	3.1
Dr. Okafor	96	\$28.3M	2.9
Dr. Grace	82	\$20.7M	3.3

Key Analytical Insights

1. **Primary Revenue Driver:** Neurology is the highest revenue-generating department, yielding \$33.3M.
2. **Top Performing Branch:** The Ibadan branch leads both in revenue generation (\$40M) and patient volume (131 patients).
3. **Primary Diagnosis Volume:** Migraine is the most frequently recorded diagnosis, accounting for 111 cases.
4. **Dominant Treatment Type:** Surgery is the most utilized treatment procedure, serving 135 patients.
5. **Highest Physician Revenue:** Dr. Ade generated the highest overall revenue at \$32.7M.
6. **Highest Physician Patient Volume:** Dr. Musa attended to the highest number of patients (113 patients).
7. **Patient Satisfaction Leader:** Dr. Grace recorded the highest patient satisfaction rating (3.3 out of 5.0) despite seeing the lowest patient volume (82).
8. **Clinical Quality Risk:** Cardiology displays the highest readmission rate (54.08%) among all departments.
9. **Financial Preferred Channel:** Bank Transfer is the primary source of revenue intake (\$52M).
10. **Demographic Dependency:** Senior citizens (61+) represent the largest patient cohort (156 patients / 31.2%).

Strategic Recommendations

- 1. Reduce Readmission Rates in High-Risk Departments:** Implement mandatory post-discharge follow-up protocols for Cardiology (54.08% readmission rate) and Orthopedics (50.00%) to optimize treatment protocols and reduce post-acute complications.
- 2. Balance Doctor Workloads to Boost Satisfaction:** Address the inverse correlation between doctor workload and patient satisfaction. Dr. Ade (111 patients, 2.8 rating) and Dr. Okafor (96 patients, 2.9 rating) should have their patient loads adjusted or supported with assistant personnel to match the higher quality-of-care scores seen under Dr. Grace (3.3 rating).
- 3. Expand Senior Care Services:** With Senior patients (61+) forming the largest demographic (156 patients), establish specialized geriatric care pathways, chronic pain management (addressing top diagnoses like Migraine and Hypertension), and streamlined admission processes.
- 4. Optimize Insurance Coverage and Partnerships:** Given that nearly half the patient base is uninsured (48.2%), form strategic partnerships with Health Maintenance Organizations (HMOs) and launch flexible installment options for self-pay transfers to mitigate default financial risk.
- 5. Standardize Best Practices from Top Branches:** Analyze operational models at the Ibadan branch (\$40M revenue, 131 patients) and scale successful patient processing strategies across lower-performing branches such as Abuja and Lagos.

Conclusion

This report provides Wellness Hospital management with a data-driven roadmap to optimize clinical operations, elevate patient care, and accelerate revenue growth. By addressing high readmissions in Cardiology (54.08%), balancing doctor workloads to improve satisfaction, and replicating the success of top revenue drivers like Neurology (\$33.3M) and the Ibadan branch (\$40M), executive management can eliminate operational inefficiencies.

Tailoring care pathways for the primary Senior demographic (156 patients) and expanding insurance partnerships to capture uninsured market share (48.2%) will stabilize cash flow, strengthen brand loyalty, and ensure long-term financial sustainability.



Thank You